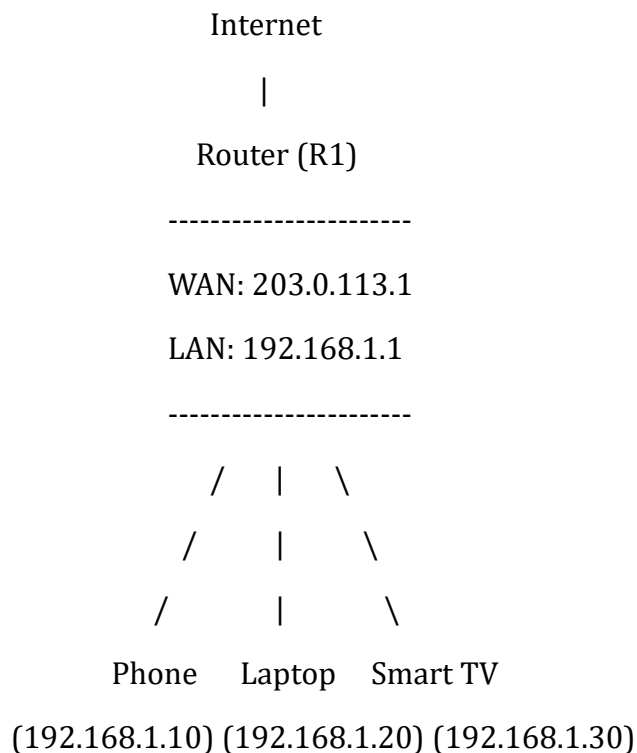


IP Routing and Routing Protocols

1. Draw a simple network diagram showing three devices (like your phone, a friend's laptop, and a smart TV) connected through a router, and label the IP addresses for each device and the router's interfaces.

ANS : Network Diagram



- **Device IP Addresses**

Device	IP Address
Router (LAN Interface)	192.168.1.1
Router (WAN Interface)	203.0.113.1
Phone	192.168.1.10
Laptop	192.168.1.20
Smart TV	192.168.1.30

2. Simulate manual IP routing by writing a step-by-step explanation of how a packet would travel from your laptop to a friend's computer in a different subnet, including which router(s) it passes through and how the routing decision is made.

Hint: Use made-up IP addresses and show the routing table entries that would be checked at each hop.

ANS :

- **Network Topology**

Laptop Router R1 Router R2 Friend's PC
 192.168.1.10 -----> 192.168.1.1/10.0.0.1 -----> 10.0.0.2/192.168.2.1 -----> 192.168.2.20

- **Device IP Address Table**

Device	IP Address
Laptop	192.168.1.10
R1 LAN	192.168.1.1
R1-R2 Link	10.0.0.1
R2-R1 Link	10.0.0.2
R2 LAN	192.168.2.1
Friend's PC	192.168.2.20

- **Step-by-Step Packet Flow**

Step 1: Laptop wants to send data to Friend's PC (192.168.2.20).

Step 2: Laptop checks destination IP.

- Destination = 192.168.2.20
- Laptop subnet = 192.168.1.0/24
- Since destination is outside local subnet, packet is sent to default gateway 192.168.1.1 (R1).

Step 3: Router R1 receives the packet.

- R1 checks its routing table.
- R1 finds route to 192.168.2.0/24 via Router R2 (10.0.0.2).
- Packet forwarded to R2.

Step 4: Router R2 receives packet.

- R2 sees destination network 192.168.2.0/24 is directly connected.
- Packet forwarded directly to Friend's PC.

Step 5: Friend's PC receives packet successfully.

3. Create a routing table for a fictional Zomato-style delivery network with three branches in different cities, each on a different subnet. Specify the network address, subnet mask, next hop, and interface for each route.

ANS :

- **Network Layout**

Branch	City	Network
Branch A	Ahmedabad	192.168.10.0/24
Branch B	Mumbai	192.168.20.0/24
Branch C	Delhi	192.168.30.0/24

- **Central Router Routing Table**

Network Address	Subnet Mask	Next Hop	Interface
192.168.10.0	255.255.255.0	Direct	G0/0
192.168.20.0	255.255.255.0	10.0.1.2	S0/0/0
192.168.30.0	255.255.255.0	10.0.2.2	S0/0/1

- **Example**

Ahmedabad orders are handled locally.

Mumbai traffic forwarded through next hop 10.0.1.2.

Delhi traffic forwarded through next hop 10.0.2.2.

4. Compare static routing and dynamic routing protocols (like RIP or OSPF) by listing 3 pros and 3 cons of each, using examples from real-world apps or services you use (e.g., how a food delivery app might benefit from dynamic routing).

ANS :

Feature	Static Routing	Dynamic Routing
Configuration	Manual	Automatic
Scalability	Small networks	Large networks
Updates	Manual	Automatic

- **Static Routing**

Advantages

1. Simple to configure in small networks.
2. No routing protocol overhead.
3. More secure because routes are manually controlled.

Disadvantages

1. Difficult to manage in large networks.
2. Network changes require manual updates.
3. No automatic failover.

- **Dynamic Routing**

Advantages

1. Automatically discovers routes.
2. Adapts to network changes.
3. Suitable for large organizations.

Disadvantages

1. More CPU and memory usage.
2. More complex configuration.
3. Routing updates consume bandwidth.

5. Use ChatGPT or another AI tool to generate a sample configuration for setting up OSPF routing between two routers, then explain in your own words what each configuration line does.

Hint: Paste the AI-generated config and annotate each part with a short description.

ANS :

- **Router R1 Configuration**

```
Router(config)# router ospf 1
```

```
Router(config-router)# network 192.168.1.0 0.0.0.255 area 0
```

```
Router(config-router)# network 10.0.0.0 0.0.0.3 area 0
```

- **Router R2 Configuration**

```
Router(config)# router ospf 1
```

```
Router(config-router)# network 192.168.2.0 0.0.0.255 area 0
```

```
Router(config-router)# network 10.0.0.0 0.0.0.3 area 0
```

- **Command Description**

Command	Description
router ospf 1	Starts OSPF process ID 1 on the router.
network 192.168.1.0 0.0.0.255 area 0	Advertises the 192.168.1.0/24 network into OSPF Area 0.
network 10.0.0.0 0.0.0.3 area 0	Advertises the link between routers into OSPF Area 0.
network 192.168.2.0 0.0.0.255 area 0	Advertises the 192.168.2.0/24 network into OSPF.
area 0	Specifies the OSPF backbone area used for routing.

- **How OSPF Works Here**

1. R1 and R2 become OSPF neighbors.
2. They exchange routing information.
3. R1 learns about network 192.168.2.0/24.
4. R2 learns about network 192.168.1.0/24.
5. Packets can now travel automatically between both networks without adding static routes.